



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 - 69



**KATO Rice Thresher/Husker-Sheller
(for Palay), Model: KMT40**

**KATO KMT40 RICE THRESHER/HUSKER SHELLER (for palay), driven by
YANMAR TF 90MLY DIESEL ENGINE**

Test Report No. : 2013 - 69
Date of Test : March 26, 2013
Test Valid Until : March 26, 2018

Tested as Rice Thresher

Brand : KATO
Model : KMT40
Type : Open, single cylinder, with sifter and fans
Manufacturer : KATO METAL CORPORATION
No. 58 E. Miranda St., Paso de Blas
Valenzuela City
Test Requested by : TRAMAT MERCANTILE, INC.
747-749 S. Padilla St., Sta. Cruz, Manila
Machine Classification : Commercial

SUMMARY

The KATO KMT40 Rice Thresher/Husker-Sheller powered by a 7.09 kW YANMAR TF 90MLY diesel engine was tested for performance using C-18 rice variety at 14.7% MC_{wb}, Grain-Straw Ratio of 0.66, and Straw Length of 510 mm. The thresher had a Threshing Capacity of 3073 kg/h with threshed grain Purity of 98.70%. At a Threshing Cylinder Speed of 760 rpm and Peripheral Speed of 1132 m/min, Threshing Recovery was 99.10% and Threshing Efficiency was 100%. The machine had a Total Threshing Loss of 0.90% consisting of 0.46% Blower Loss, 0.44% Separation Loss, no Unthreshed Loss and no Scattering Loss.

For purposes of standardization, the average Corrected Threshing Capacity was 2694 kg/h based on 20% MC_{wb}, 0.55 Grain-Straw Ratio and 100% Purity.

Grain quality analysis of threshed grains showed a net percent Cracked Grain of 1.68 % and 0.28 Broken Grains.

The Fuel Consumption Rate of the 7.09 kW YANMAR TF 90MLY diesel engine was 2.48 L/h.

During the test, six persons were needed to operate the thresher: two for hauling of test materials to the thresher; one for feeding of test materials; two for collecting and bagging of threshed grains; and one, for recycling of uncleaned grains and tailings.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance using a 7.09 kW YANMAR TF 90MLY diesel engine as its primemover. The test was conducted in Brgy. San Pedro, Morong, Rizal on March 26, 2013 and was evaluated on the following items:

1. Threshing Capacity
2. Quality of Work
 - a. Purity of Threshed Grain
 - b. Percent Cracked Grain
 - c. Percent Broken Grain
3. Threshing Recovery
4. Threshing Efficiency
5. Threshing Losses
 - a. Blower Loss
 - b. Separation Loss
 - c. Unthreshed Loss
 - d. Scattering Loss

METHOD OF TEST

The thresher was tested based on the Philippine Agricultural Engineering Standard “PAES 205:2000 Agricultural Machinery – Mechanical Rice Thresher - Methods of Test”.

OPERATOR'S MANUAL

No operator's manual nor brochure of the thresher was submitted.

DESCRIPTION OF THE MACHINE

The KATO KMT40 Thresher as shown in **Figure 1**, was basically an axial flow type mechanical paddy thresher with cleaning components. It was powered by a 7.09 kW YANMAR TF 90MLY diesel engine which drove directly the fan shaft through a double groove B-section pulley and V-belt which in turn drove the threshing cylinder shaft through a double groove B-section pulley and V-belt. The fan shaft also drove the oscillating screen shaft through a separate single groove B-section pulley and V-belt. Belt-pulley power transmission system of the rice thresher is shown in **Figure 2**.

The machine consisted of the following main components: feeding table, threshing assembly and cleaning section and discharge section. The folding type feeding table (**Fig. 3**) was attached to the main frame through bolts and nuts. Feeding was controlled manually.

The threshing assembly (**Fig. 4**) consisted of a single, open type cylinder and a lower concave grill. The threshing cylinder had a length of 755 mm and a diameter of 475 mm and had four longitudinal anchor bars. Three anchor bars were welded with six pegteeth while one anchor bar was welded with seven pegteeth. The pegteeth were arranged offset with respect to the pegteeth in the next anchor bar. The lower concave grills were made of 8 mm diameter mild steel. The threshing section was provided with a top cover hinged to the main frame. The top cover was welded with five spiral louvers.

The cleaning assembly consisted of two straight radial type centrifugal fans (**Fig. 5**) which separated the straw chaffs and other light impurities from the threshed grain, and a perforated (14 mm) oscillating screen (**Fig. 6**). Clean threshed grain was conveyed and collected through an inclined chute. Uncleaned grains and tailings were likewise collected by an inclined chute and manually fed and recycled back to the threshing cylinder.

The main frame of the machine was made of angle bars. A chassis was also incorporated where transport wheels and its assembly were attached. The machine was also provided with two detachable towing bars and hitch point.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. KATO Rice Thresher/Husker-Sheller
Model: KMT40



Figure 2. Belt-Pulley Power Transmission Systems

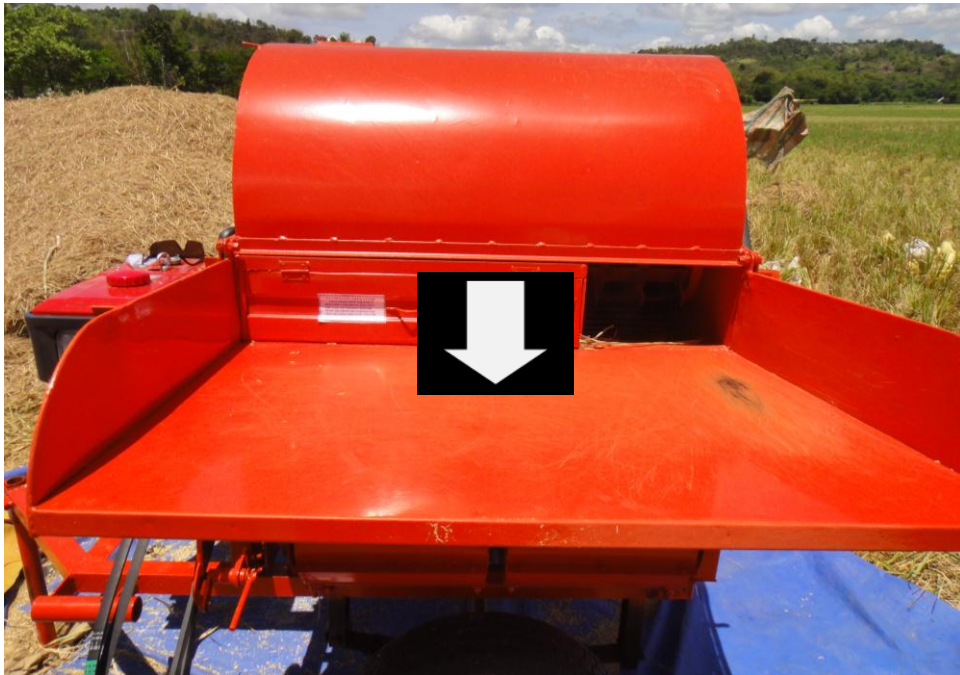


Figure 3. Feeding Table

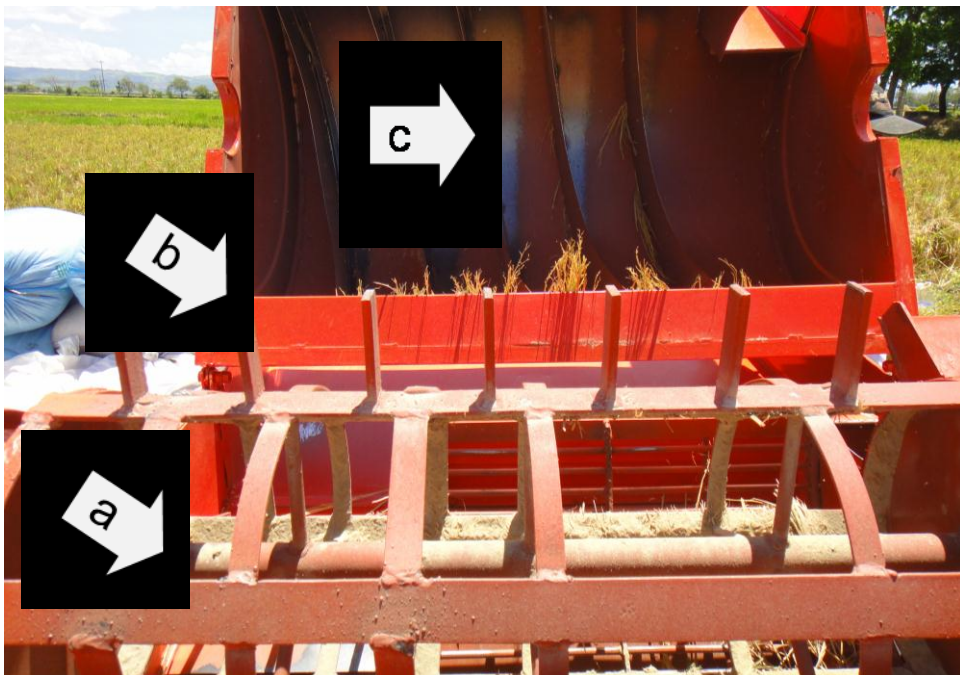


Figure 4. Threshing Assembly
a) Threshing cylinder
b) Pegteeth
c) Louver

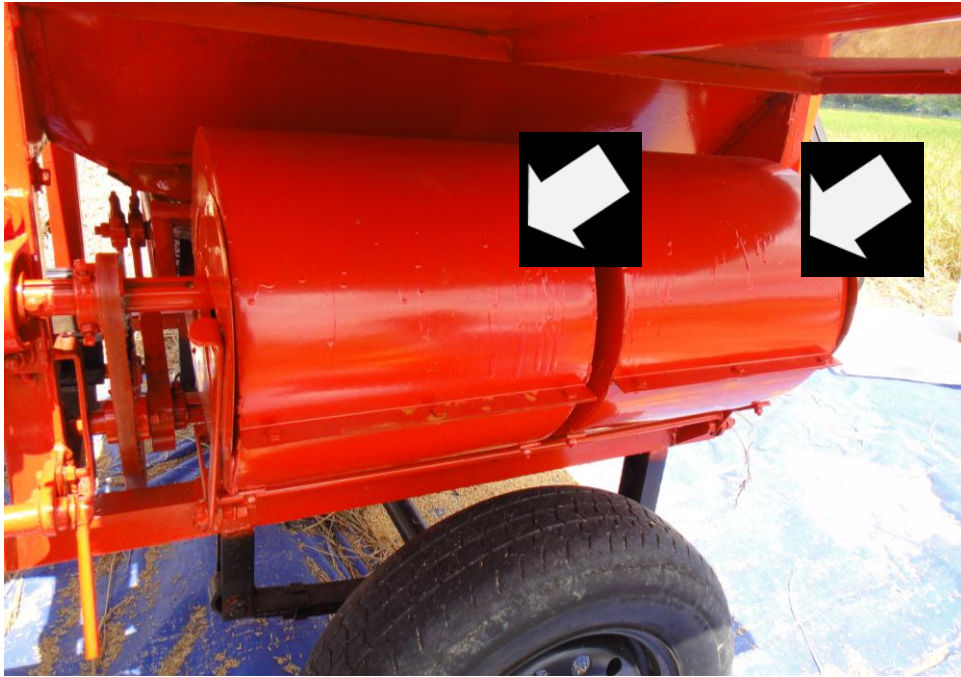


Figure 5. Straight Radial Centrifugal Fans



Figure 6. Oscillating Screen

SPECIFICATIONS

Table 1. Detailed specifications of the KATO KMT40 Rice Thresher/Husker-Sheller

A.	Brand	:	KATO
B.	Model	:	KMT40
C.	Serial No.	:	-
D.	Manufacturer	:	KATO METAL CORP. Valenzuela City
E.	Overall Dimensions (Transport mode) and Weight		
1.	Length (mm)	:	1880
2.	Width (mm)	:	1600
3.	Height (mm)	:	1860
4.	Weight (kg) without engine	:	No Data
F.	Average component Shaft Speeds (with load)		
1.	Threshing cylinder (rpm)	:	760
2.	Fan (rpm)	:	1722
3.	Oscillating screen (rpm)	:	749
G.	Threshing cylinder		
1.	Type	:	Open cylinder
2.	Size, L x D (mm)	:	755 x 475
3.	Cylinder teeth		
a)	Type	:	Knife Bar
b)	Size, L x W x T (mm)	:	85 x 25 x 6
c)	No. per anchor bar	:	6, 6, 6, 7
d)	Means of attachment	:	Welded
e)	Arrangement	:	Offset with respect to the pegteeth in the next anchor bar
f)	Material	:	Mild Steel

Table 1, Detailed specifications of the KATO KMT40 Rice Thresher (Cont'n)

	4.	No. of anchor bars	:	4
	5.	Cylinder cover material	:	B.I. Sheet, Ga. #16
H.		Louvers		
	1.	Material	:	B.I. Sheet, Ga. #16
	2.	Number	:	5
I.		Lower concave		
	1.	Grills	:	Mild steel, 8 mm diameter
	2.	Spacing between grills (mm)	:	19
	3.	Clearance between concave and pegteeth (mm)	:	20
J.		Fan		
	1.	Type	:	Radial centrifugal
	2.	No. of unit	:	2
	3.	Length (mm)	:	325 (per unit)
	4.	Number of blades	:	4
	5.	Material	:	C.R Sheet, Ga. #20 (blade and housing)
K.		Oscillating Screen		
	1.	Dimensions, L x W (mm)	:	900 x 710
	2.	Material	:	Perforated Screen (C.R. Sheet, Ga. #18)
	3.	Size of Perforations (mm)	:	12
	4.	Stroke (mm)	:	12
L.		Feeding Table		
	1.	Dimensions, L x W (mm)	:	1010 x 635
	2.	Height from the ground (mm)	:	1100
	3.	Mode of attachment	:	Bolted
	4.	Material	:	C.R Sheet, Ga. #20

Table 1. Detailed specifications of the KATO KMT40 Rice Thresher (Cont'n)

M.	Main Frame	:	Angle bar (51 mm x 51 mm x 5 mm)
N.	Power Transmission System		
1.	Engine to Fan	:	Double groove, B-section pulley and V-belt
	Engine pulley	:	152 mm x 2B x 32 mm
	Fan pulley	:	203 mm x 2B x 32 mm
	Belt	:	B-57
2.	Fan to Oscillating Screen shaft	:	Single groove, B-section pulley and V-belt
	Fan pulley	:	76 mm x 1B x 32 mm
	Oscillating Screen pulley	:	178 mm x 1B x 25 mm
	Belt	:	B-36
3.	Fan to Threshing Cylinder	:	Double groove, B-section pulley and V-belt
	Fan pulley	:	127 mm x 2B x 32 mm
	Threshing Cylinder pulley	:	254 mm x 2B x 25 mm
	Belt	:	B-67
O.	Transport Device	:	Rubber tires
P.	Discharge device	:	Inclined chute
Q.	Engine (Prime mover)		
1.	Brand	:	YANMAR
2.	Model	:	TF 90MLY
3.	Make	:	YANMAR CO. LTD., Indonesia
4.	Rated Maximum Power (kW)	:	7.09 @ 2400 rpm
5.	Rated Continuous Power (kW)	:	6.34 @ 2400 rpm
6.	Type	:	Four-stroke, water-cooled single horizontal cylinder diesel engine
7.	Starting System	:	Manual cranking
8.	Cooling System	:	Water-cooled
9.	Weight (kg)	:	100.0

RESULTS AND DISCUSSIONS

Results of field performance tests are shown in **Table 2 - 5**. Refer to **Annex 1** for formula used in the computation of results.

Threshing Capacity

Test results showed that the machine had a Threshing Capacity of 3073 kg/h using C-18 rice variety at 14.7% MC_{wb}, Grain-Staw Ratio of 0.66, and Straw Length of 510 mm.

For purposes of standardization, the Corrected Threshing Capacity based on 20% MC_{wb}, 0.55 Grain-Staw Ratio, and 100% Purity, was 2694 kg/h.

Performance

The performance of the thresher was evaluated in terms of its Threshing Recovery, Threshing Efficiency, Threshing Losses, Purity and Grain Quality.

Threshing Efficiency and Recovery. The Threshing Efficiency of the machine was 100% and the Threshing Recovery was 99.10%

Threshing Loss. Total Threshing Loss was 0.90% consisting of 0.46% Blower Loss, 0.44% Separation Loss, no Unthreshed Loss and no Scattering Loss.

Purity. The Purity of collected threshed grain was 98.70%.

Grain quality. Quality of threshed grain was determined in terms of percent cracked and broken grains. Results showed that the threshed output had a net percentage Cracked Grains of 1.68% and 0.28% Broken Grains. (Cracked grains referred to those grains whose endosperms were partially fractured while broken grains referred to those grains that were split into 2 or more pieces.)

Fuel Consumption

During testing, the 7.09 kW YANMAR TF 90MLY diesel engine had a Fuel Consumption Rate of 2.48 L/h.

Labor Requirement

Six persons were needed to operate the machine (**Fig. 7**): two for hauling of test material to the thresher; one for feeding; two for collecting and bagging of threshed grains; and, one for recycling of uncleaned grains and tailings.

Observations:

1. The thresher's belt power transmission was provided with belt guard (**Fig. 8**).
2. The thresher was provided with safety stickers (**Fig. 9**).



Figure 7. Operators of the thresher



Figure 8. Belt guard



Figure 9. Sticker attached to the thresher

PERFORMANCE TEST

Machine Tested: KATO Rice Thresher/Husker-Shellor (for palay)

Model: KMT40

powered by

7.09 kW YANMAR Diesel Engine

Model: TF 90MLY

Location of Test: Brgy. San Pedro

Morong, Rizal

Date of Test : March 26, 2013

Table 2. Condition of crop used during the test

ITEMS	Data
1. Rice variety	C-18
2. Days after harvest	3
3. Straw length (mm)	510
4. Grain-Straw Ratio	0.66
5. Grain moisture content (%), w.b.	14.7

Table 3. Speed of thresher components during the test

ITEMS	Data
1. Engine (rpm)	
Without load	2509
With load	2338
2. Threshing Cylinder (rpm)	
Without load	950
With load	760
3. Threshing Cylinder Peripheral Speed (m/min)	
Without load	1416
With load	1132
4. Fan (rpm)	
Without load	2007
With load	1722
5. Oscillating Screen (rpm)	
Without load	828
With load	749

Table 4. Thresher performance during the test

ITEMS	Data
1. Thresher capacity (kg/h)	
Actual	3073
Corrected*	2694
2. Purity of threshed grain (%)	98.70
3. Threshing recovery (%)	99.10
4. Threshing efficiency (%)	100
5. Blower loss (%)	0.46
6. Separation loss (%)	0.44
7. Unthreshed loss (%)	Nil
8. Scattering loss (%)	Nil
9. Total threshing loss (%)	0.90
10. Fan air velocity (m/s)	
Without load	11.2
11. Noise level [db(A)]	
a. Bagger's ear level	
Without load	89.2
With load	92.4
b. Feeder's ear level	
Without load	86.2
With load	89.2
12. Primemover's fuel consumption (YANMAR TF 90MLY Diesel Engine), L/h	2.48

* Adjusted to 20% grain moisture content, 0.55 grain-straw ratio and 100% purity of threshed grain

Table 5. Laboratory grain analysis of samples gathered during testing

ITEMS	Data
1. Before Threshing	
Cracked	1.66
Broken	0.10
2. After Threshing	
Cracked	3.34
Broken	0.38
3. Net Percent	
Cracked	1.68
Broken	0.28

DATA COMPUTATION

ANNEX 1. Formula used during calculations and testing

1. Grain-Straw Ratio (R)

$$R = \frac{W_g}{W_s} \quad \text{where:}$$

W_g = weight of grains (g)

W_s = weight of sample (grain and straw), g

2. Fuel consumption (F_C)

$$F_C = \frac{F_l}{T_o} \quad \text{where:}$$

F_C = Fuel consumption (L/h)

F_l = Total amount of fuel consumed (L)

T_o = Time of operation (h)

3. Actual Threshing Capacity (C_a)

$$C_a = \frac{W_c}{T_o} \quad \text{where:}$$

C_a = Actual threshing capacity (kg/h)

W_c = Weight of threshing output (kg)

T_o = Duration of operation (h)

4. Corrected Threshing Capacity (C_C)

$$C_C = \frac{100 - MC_o}{100 - MC_m} \times \frac{R_m}{R_o} \times P \times C_a \quad \text{where:}$$

C_C = Corrected threshing capacity (kg/h)

MC_o = Observed grain moisture content (%)

MC_m = Grain moisture content, at 20%

P = Purity (%)

R_m = Standard grain-straw ratio of 0.55

R_o = Observed grain-straw ratio

5. Purity (P)

$$P = 100\% - \% \text{Impurity}$$

6. Impurity (I)

$$I = \frac{G_{US} - G_{CS}}{G_{US}} \quad \text{where:}$$

I = Impurity (%)

G_{US} = Uncleaned grain sample (g)

G_{CS} = Cleaned grain sample (g)

7. Threshing Efficiency (T_e)

$$T_e = \frac{W_c + S_l + B_l}{W_c + S_l + B_l + U_l} \times 100 \quad \text{where:}$$

T_e = Threshing Efficiency (%)

S_l = Separation loss (kg)

B_l = Blower loss (kg)

U_l = Unthreshed loss (kg)

8. Threshing Recovery (T_r)

$$T_r = \frac{W_c}{W_c + S_l + B_l + U_l} \times 100$$

9. Percent Unthreshed loss (U_l)

$$U_l = \frac{U_l}{W_c + S_l + B_l + U_l} \times 100$$

10. Percent Separation loss (S_l)

$$S_l = \frac{S_l}{W_c + S_l + B_l + U_l} \times 100$$

11. Percent Blower loss (B_l)

$$B_l = \frac{B_l}{W_c + S_l + B_l + U_l} \times 100$$

12. Cracked grain (C_g)

$$C_g = \frac{N_{Cg}}{100 \text{ grain sample}} \times 100 \quad \text{where:}$$

C_g = Percent cracked grain

N_{Cg} = Number of cracked grain

13. Broken grain (B_g)

$$B_g = \frac{N_{Bg}}{100 \text{ gram sample}} \times 100 \quad \text{where:}$$

B_g = Percent broken grain

N_{Bg} = Weight of broken (g)

14. Net Cracked Grains

$$NG_C = IG_C - FG_C \quad \text{where;}$$

NG_C = Net Cracked Grains

IG_C = Number of Cracked grains before threshing

FG_C = Number of Cracked grains after threshing

15. Net Broken Grains

$$NG_B = IG_B - FG_B \quad \text{where;}$$

NG_B = Net Broken Grains

IG_B = Weight of Broken Grains before threshing

FG_B = Weight of Broken Grains after threshing

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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